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Performance evaluation of continuous $PM_{2.5}$ mass concentration monitors

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Abstract

$PM_{2.5}$ mass concentrations were measured on a short time interval basis using the continuous ambient mass monitor (CAMM) and real-time ambient mass sampler (RAMS) along with a differential tapered element oscillating microbalance (TEOM) in Rubidoux CA, and intercomparisons were made among these samplers to evaluate their measurement performance of $PM_{2.5}$ mass. Nephelometers were used as an additional sampler to assist in the between-sampler comparisons. Based on the correlation with particulate nitrate concentrations and particulate mass, the differential TEOM showed better measurement of semivolatile species in $PM_{2.5}$ than either the CAMM or the RAMS. The RAMS typically measured more mass than the CAMM. The results suggest that the differential TEOM was better reflecting the $PM_{2.5}$ mass concentrations, with less impact from the loss of semivolatile components. Particulate ammonium nitrate losses estimated from the paired Harvard-EPA annular denuder system and a commercial federal reference method (FRM) sampler were equivalent to the $PM_{2.5}$ mass difference between the paired differential TEOM and FRM samplers. It suggests that volatilization of semivolatile ammonium nitrate was substantial at Rubidoux, and thus, the differential TEOM performed better in measurement of the actual $PM_{2.5}$

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mass concentrations, including semivolatile and non-volatile PM, with less loss of semivolatile materials from the filter.
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Keywords: $PM_{2.5}$; Mass; Measurement; Continuous ambient mass monitor; Real-time ambient mass sampler; Differential tapered element oscillating microbalance; Federal reference method

1. Introduction

Reliable measurement of airborne particulate matter (PM) mass is important in assessing the roles of PM in epidemiological studies as positive associations have been reported between PM concentration levels and adverse respiratory effects (Dockery et al., 1993; Gordian, Ozkaynak, Xue, Morris, & Spengler, 1996). A number of PM measurement methods are available from the conventional filter sampling system to the semi-continuous mass monitors, with and without a gas/vapor denuding system. The most frequently used method for PM collection is to collect particles on filter for a certain amount of time, usually 24 h, and determine the net mass of particles by weighing the filter before and after sampling. Typically, some type of equilibration process at a low relative humidity and standard temperature is used to stabilize the filter weight. This process also reduces the amount of particle-bound water.

The United States Environmental Protection Agency (US EPA) regulations require the use of the federal reference method (FRM), or an equivalent method for PM mass measurement. The FRM is a filter-based, 24-h sampling technique with a pre- and post-weighing of the filter, prepared under standardized laboratory conditions. However, this method has some drawbacks, including the ill-defined, non-quantifiable mass changes of PM on the filter because of volatilization of semi-volatile materials, adsorption/desorption of gaseous species, and chemical reactions of gases with collected particles (Patashnick, Rupprecht, Ambs, & Meyer, 2001). To prevent mass changes on the filter and provide more reliable airborne PM mass concentration, a variety of continuous PM mass measurement systems have been employed to provide higher time resolution, real-time measurements and to reduce personnel time required to change and weigh filters. Such techniques including the continuous ambient mass monitor (CAMM; Babich, Davey, Allen, & Koutrakis, 2000), the real-time ambient mass sampler (RAMS; Eatough, Obedi, Pang, Ding, Eatough, & Wilson, 1999; Eatough, Eatough, Obedi, Pang, Modey, & Long, 2001; Pang, Eatough, Modey, & Eatough, 2002), and the differential tapered element oscillating microbalance (differential TEOM; Patashnick et al., 2001), were used to measure $PM_{2.5}$ mass concentrations in Rubidoux, CA. These samplers can measure $PM_{2.5}$ mass concentrations at time intervals of less than 1 h, during which the aerosol composition may remain near equilibrium.

In this study, the performance of these continuous samplers was evaluated. Comparisons were made among $PM_{2.5}$ mass concentrations determined from these continuous PM mass monitors and also between continuous and filter-based FRM $PM_{2.5}$ mass concentrations. Along with PM mass, concentrations of semivolatile species were measured semi-continuously, and the impact of semivolatile species on mass changes of $PM_{2.5}$ in the air was also examined.

2. Measurements

Various continuous particle mass monitoring systems were deployed in Rubidoux, CA to measure mass concentrations of $PM_{2.5}$ from 15 August through 7 September, 2001. These measurements were conducted

in conjunction with the Southern California Particle Center and Supersite (SCPCS) measurement and monitoring program. The measurements were made within the SCPCS particle instrumentation unit located at Rubidoux, CA (33.9996N, 117.4161W, elevation 248.4 m). Rubidoux is located downwind of the major nitrogen oxides (NO_x) emissions sources (i.e., traffic vehicles) of the Los Angeles basin and is susceptible to an aerosol plume after aging for several hours to a day (Fine, Jaques, Hering, & Sioutas, 2003). Rubidoux is also downwind of significant ammonia emission sources from the nearby farmlands and dairy areas in western Riverside and San Bernardino Counties that results in high concentrations of ammonium nitrate (Christoforou, Salmon, Hannigan, Solomon, & Cass, 2000). This situation is atypical of most of the United States, but provides a severe test of mass measurement systems in terms of their ability to properly measure the particulate nitrate as well as deal with a significant amount of semivolatile organic matter.

$PM_{2.5}$ mass concentrations were measured on a 5-min to 1-h basis using the RAMS, the CAMM, and the differential TEOM monitor. The observed $PM_{2.5}$ mass concentrations were averaged to 1 h intervals in order to compare one monitor to another. Along with these continuous samplers, two nephelometers equipped with or without a Nafion dryer were collocated to measure scattering by particles. A 35 °C TEOM with a Nafion dryer was also deployed in this study, but the results were not used because the unit malfunctioned during the study.

The RAMS measures $PM_{2.5}$ mass concentrations based on dual diffusion denuder technology and a particle concentrator combined with dual TEOMs (Eatough et al., 1999, 2001; Pang et al., 2002). The CAMM uses the change in pressure differential across the filter between pre- and post-sampling to estimate the $PM_{2.5}$ loading (Babich et al., 2000).

The differential TEOM monitor (Patashnick et al., 2001; Jaques, Ambs, Grant, & Sioutas, 2004) consists of a size selective inlet, a diffusion dryer, and matched pairs of electrostatic precipitators (ESPs) and TEOMs. This instrument is based on the direct real-time mass measurement capability of the TEOM monitor. Downstream from a common size selective inlet and upstream of the dual TEOM mass sensors are ESPs. The ESPs are alternatively switched on and off for 5 min and out of phase with each other. Frequency data are collected for both TEOM sensors on a continuous basis. During the period when the ESP is discharged (i.e., switched off), the TEOM measures the PM in a manner analogous to the TEOM with an SES dryer (Meyer, Patashnick, Ambs, & Rupprecht, 2000). During the period when the ESP is energized (i.e., switched on), the PM is removed from the sample stream and retained inside of the ESP (Jaques et al., 2004). While the ESP is energized, volatilization of previously collected particles or adsorption of gaseous species from the sample stream may occur. The mass change of the filter with the ESP energized is subtracted from the mass change of the filter during the ESP discharged to provide an artifact-corrected net mass measurement (Hering et al., 2004).

Along with these continuous PM mass samplers, two nephelometers (Radiance Research M903) were deployed to measure scattering by particles: one was equipped with a Nafion dryer and the other was not.

In addition, semi-continuous measurements of semivolatile fine PM species, organic carbon and nitrate, were made using a Sunset Laboratory Semi-Continuous OC-EC Carbon Aerosol Analyzer (Sunset Labs, Forest Grove OR) and an integrated collection and vaporization cell (ICVC) system, respectively. Organic carbon (OC) and elemental carbon (EC) concentrations were determined every hour. The ICVC system (Aerosol Dynamics Inc., Berkeley CA) uses 3 cascade cells to provide automated (10 min resolution) monitoring of size-segregated (2.5–1.0, 1.0–0.45, and 0.45–0.1 μm) fine particulate nitrate (Fine et al., 2003). Concentrations of size-fractioned particle nitrate were determined from each of 3 stages and combined to provide total nitrate (i.e., $PM_{2.5}$ nitrate) concentrations.

Time integrated $PM_{2.5}$ mass concentrations were determined using a partisol-Plus FRM sampler (R&P Model 2025) every 3 days, following the South Coast Air Quality Management District's schedule. The PM mass samples were collected during 4 periods of each day (6–10am, 10am–4pm, 4–8pm, and 8pm–6am). Particulate nitrate was determined from the FRM Teflon filter after PM mass was weighed. A Harvard-EPA annular denuder system (HEADS) was also used to collect artifact-free $PM_{2.5}$ particulate nitrate on a Nylon filter for the same time periods as the partisol FRM. The HEADS used a carbonate-carbon denuder to remove nitric acid, followed by a Teflon filter and two sodium carbonate filters to collect semivolatile particle nitrate and to correct for positive artifacts from nitrogen dioxide (Fine et al., 2003). It was used as the measure of artifact-free $PM_{2.5}$ particulate nitrate.

3. Results and discussion

Fig. 1 presents hourly $PM_{2.5}$ mass concentrations measured using the continuous CAMM, RAMS, and differential TEOM monitors and scattering by particles measured using the nephelometers as well as hourly concentrations of $PM_{2.5}$ OC from the Sunset Laboratory OC/EC analyzer and nitrate from the ICVC system at Rubidoux, CA. After data validation, there were many $PM_{2.5}$ mass data points missing in August. There were problems of inadequate temperature control within the trailer that resulted in loss of temperature control in several cases. There were also some problems with the air drying system that

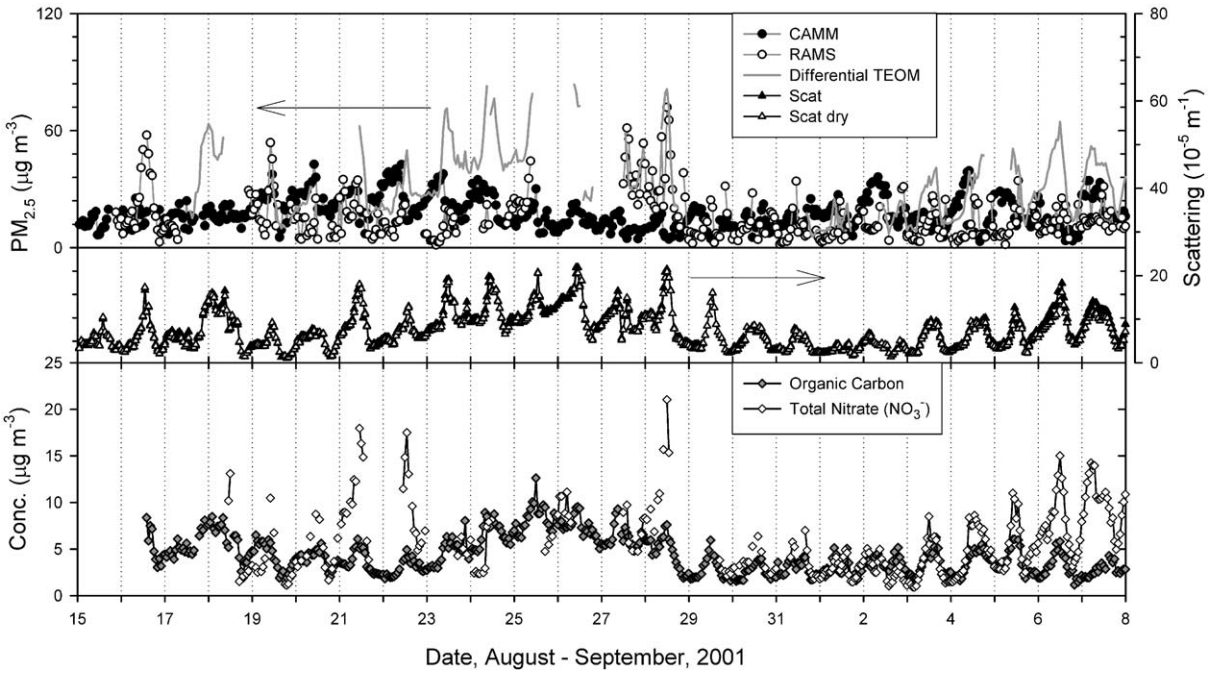


Fig. 1. Time series of hourly $PM_{2.5}$ mass concentrations in $\mu g/m^3$ measured using various continuous particle monitoring methods in Rubidoux, CA. Scattering by particles from nephelometers are also presented along with hourly concentrations of the semi-continuous particulate organic carbon and nitrate.

provides the sheath air for the RAMS. Careful inspection of the data logs from the instruments were used to evaluate the data. However, better data capture rates were achieved in September.

The CAMM showed relatively good continuity of data over the whole sampling period. The greater variation of $PM_{2.5}$ mass concentrations appears in August than in September in most of the continuous samplers, except in the CAMM which showed a lower variability with time. Among the continuous PM samplers, the RAMS and the differential TEOM show a relatively similar pattern of concentration variation. Several peaks are well matched in both samplers. However, there is a period of 23–24 August in which the differential TEOM gave values much higher than the other systems. The reason for this behavior is not known. The CAMM and/or the RAMS PM mass could potentially underestimate the mass, the differential TEOM might overestimate the mass concentration, and/or the NO_3^- also could be underestimated. The nitrate instrument has a minimum cut-point of 0.18 μm in the smallest stage and thus, may not collect all of the accumulation mode nitrate. Recent studies in Pittsburgh have suggested that the flash vaporization method tends to underestimate the NO_3^- concentrations as measured with integrated samplers (Wittig et al., 2004).

In the comparison with particle scattering from the nephelometers, it can be seen that the concentration variations of the RAMS and the differential TEOM also correspond well to the variation of light scattering in the nephelometers. This good agreement is clearer in the differential TEOM. The RAMS and the differential TEOM had similar performances in measuring $PM_{2.5}$ mass concentrations in this study, but had somewhat dissimilar performance when compared with the CAMM. Both the RAMS and the differential TEOM have filter systems to prevent the loss of semivolatile $PM_{2.5}$ species from the filter during sampling, but the CAMM depends on an assumption of low volatility loss during short measurement time intervals. This hypothesis is further supported by the observed variation of $PM_{2.5}$, OC, and nitrate concentrations shown in Fig. 1. At times of peaks for the RAMS, the differential TEOM and the nephelometers, OC and nitrate concentrations also increased. This pattern is clearly observed in nitrate and the differential TEOM measurements although many samples are missing in the nitrate data. These results suggest that the RAMS and the differential TEOM capture semivolatile and nonvolatile PM materials with lower losses than the CAMM, and second, that OC and nitrate contribute to the $PM_{2.5}$ mass concentrations in the study area. Previous studies have reported that nitrate (e.g. ammonium nitrate) constitutes a major portion of $PM_{2.5}$ mass concentrations in southern California (Christoforou et al., 2000) and represents a significant fraction of the losses in the differential TEOM (Jaques et al., 2004).

It is interesting that OC and nitrate show similar temporal profiles in Fig. 1. In particular, a strong correlation ($r^2 = 0.86$) was observed between the OC and nitrate concentrations during 2 September through 5, possibly suggesting the formation of OC by secondary processes. In that period, the OC showed a very weak correlation ($r^2 = 0.25$) with elemental carbon, which is a primary tracer of vehicle emission sources. The average (± standard deviation) ratio of OC to EC, calculated from the hourly concentrations during that period, was about 5.3 ± 3.8.

Table 1 shows average concentrations of hourly $PM_{2.5}$ mass for the whole sampling period along with hourly fine particulate nitrate and OC concentrations. The average $PM_{2.5}$ mass concentrations of the CAMM and RAMS were comparable to each other. The differential TEOM had a $PM_{2.5}$ mass concentration of 33.1 $\mu g/m^3$ on average, which is about two times higher than concentrations of the CAMM and RAMS. The average concentration of particulate nitrate for 297 valid samples was 7.22 $\mu g/m^3$, as ammonium nitrate, and contributes 40–45% to $PM_{2.5}$ mass concentrations of the CAMM and RAMS, and about 22% to the differential TEOM PM mass concentrations. The average fine particle organic aerosol concentration for 526 valid samples was 6.02 $\mu g/m^3$, multiplied by a factor of 1.4 to OC to compensate

Table 1
Hourly average concentrations ($\mu g/m^3$) of $PM_{2.5}$ mass measured using continuous PM monitors and of $PM_{2.5}$ nitrate and organic carbon determined from the ICVC system and the OC/EC analyzer, respectively

	CAMM	RAMS	Differential TEOM	Nitrate (NH_4NO_3)[a]	OC (Organics)[b]
No. of samples	533	366	313	297	526
Average ($\mu g/m^3$)	17.7	15.7	33.1	5.6 (7.22)	4.3 (6.02)

[a]Ammonium nitrate is calculated by multiplying a factor of 1.29 to particulate nitrate.
[b]Organic aerosol multiplied by a factor of 1.4 to organic carbon.

for O and H associated with organic matter, and contributes about 34%, 38%, and 18% to the PM mass of the CAMM, the RAMS, and the differential TEOM, respectively. Higher fractions would be found if a higher OC multiplier were employed as has been suggested by Turpin and Lim (2001).

3.1. Instruments comparison

Fig. 2 depicts the results of intercomparison of the continuous PM samplers and nephelometers. The descriptive information of the correlation is also given with box plots of concentrations from each instrument. In each box plot are shown percentiles, the median (solid line), and the mean (short dash line) of concentrations. There is a good correlation ($r^2 = 0.92$) between $PM_{2.5}$ mass concentrations measured by the individual TEOMs in the differential TEOM. The linear fit equation has a slope of 1.01 with an uncertainty (i.e., standard error) in parentheses of ± 0.01 and an intercept of 1.85 (± 0.54) $\mu g/m^3$. Thus, there was no significant bias between these two TEOM monitors, and PM mass concentrations from either system A or B can be used.

In the relationship between the differential TEOM and the other continuous PM samplers, a relatively weak correlation ($r^2 = 0.23$) was found with the RAMS. Their relationship is depicted as lower slope than unity, but high intercept, meaning not only that higher PM mass concentrations were measured with the differential TEOM, but also that both samplers showed dissimilar performances in measuring $PM_{2.5}$ mass concentrations. Looking at Fig. 2, however, the two monitors appear to agree well at higher $PM_{2.5}$ levels, whereas at lower concentrations the differential TEOM read about twice that of the RAMS. It should be noted that for higher PM concentrations ($> 30\ \mu g/m^3$), a strong association ($r^2 = 0.82$) was observed between the differential TEOM and RAMS. Among valid samples of both samplers given in Table 1, only the 11 paired differential TEOM-RAMS samples were available, by removing a few outliers, and the average mass concentration was $49.9\ \mu g/m^3$ for the differential TEOM and $48.5\ \mu g/m^3$ for the RAMS. These results may result from the particle collection efficiency of the concentrator used in the RAMS. At receptor sites in Los Angeles, the particle size increases with $PM_{2.5}$ levels (Kim, Shen, Sioutas, Zhu, & Hinds, 2002; Fine, Shen, & Sioutas, 2004). The larger the size, the more efficiently it is concentrated. This might explain that both samplers showed a strong correspondence at higher $PM_{2.5}$ levels. No meaningful relationship appeared either between the differential TEOM and the CAMM $PM_{2.5}$ mass concentrations ($r^2 = 0.05$), or between the RAMS and the CAMM ($r^2 = 0.07$).

A comparison was made between light scattering by particles using the non-dryer nephelometer and $PM_{2.5}$ mass concentrations by the continuous PM monitors in order to further examine the measurement performances of the PM samplers. Light scattering shows good correlation, r^2 equal to 0.84, with the

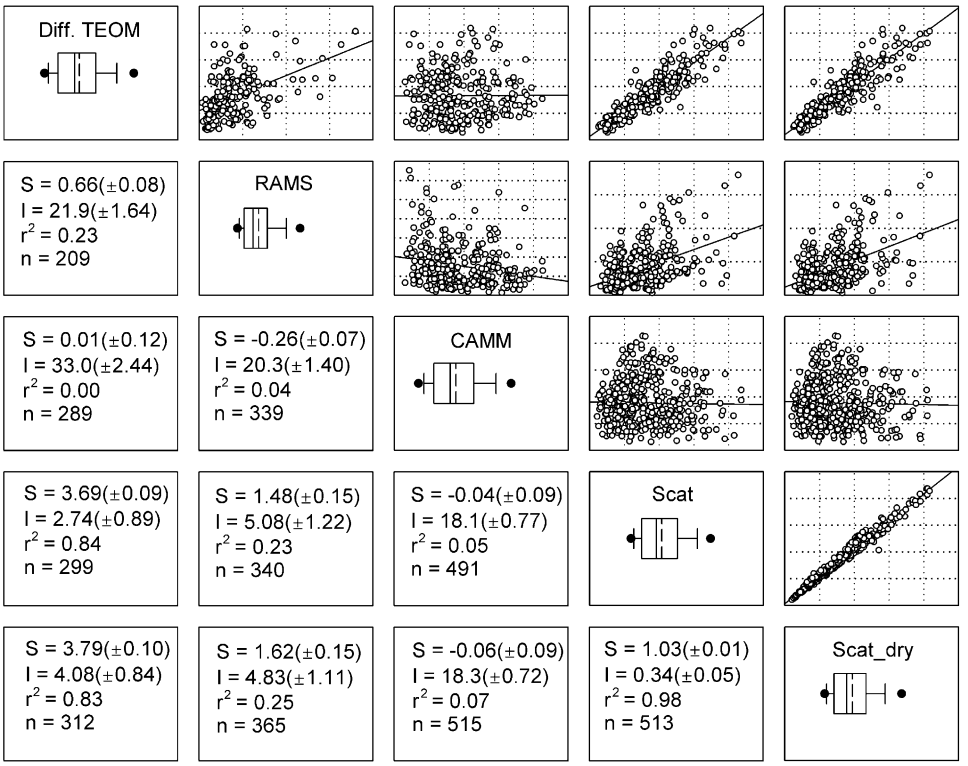


Fig. 2. Matrix scatter plots that show the linear relationship between continuous $PM_{2.5}$ mass concentrations and particle scattering. S and I indicate the slope and intercept with involved uncertainties of individual linear equation curve related, respectively. The boundary of the box closest to and farthest from zero indicates the 25th percentile and the 75th percentile, respectively. Error bars indicate the 10th and 90th percentiles. Lines within the box mark the median (solid) and the mean (short dash). Outliers (closed circle) are the 5th and 95th percentiles.

differential TEOM $PM_{2.5}$ mass concentrations. On the other hand, the correlation with the RAMS is weaker than that with the differential TEOM, and no significant correlation was observed with the CAMM. These results may suggest that, assuming little error in nephelometer measurement, the differential TEOM had better performance in the measurement of $PM_{2.5}$ mass concentrations than either the RAMS or the CAMM in the Rubidoux field study. The CAMM showed the worst performance in this comparison. The linear fit equation between particle scattering measurements with and without a Nafion dryer is Scat $= 1.03(\pm 0.01)$ Scat_dry $+ 0.34(\pm 0.05)$, with a high correlation ($r^2 = 0.98$) between the two nephelometers, showing no operational error and low impact of water vapor on particles captured, and further supporting the previous discussion.

As a metric to present measurement performances of the continuous PM mass samplers, the coefficient of divergence (COD; Wongphatarakul, Friedlander, & Pinto, 1998) was calculated pairwise between the PM samplers using the mean concentrations given in Table 1. The value of COD calculated is smallest (0.061) between the CAMM and RAMS, and increases to 0.303 when the CAMM is compared with the differential TEOM. Because the COD is calculated based on mean concentrations of PM mass samples of two samplers, a higher difference in mass concentrations causes a higher value of COD.

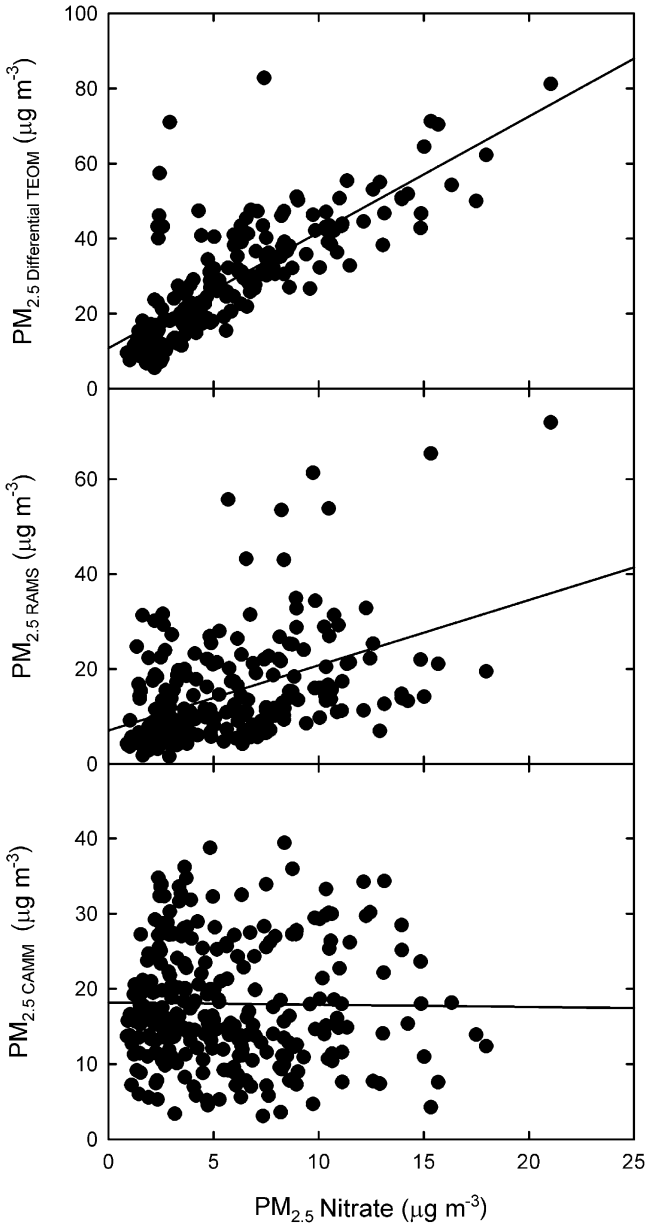


Fig. 3. Correlation between the ICVC fine particle nitrate concentrations and $PM_{2.5}$ mass concentrations determined from the differential TEOM, the RAMS, and the CAMM.

Figs. 3 and 4 show better performance of the differential TEOM in measuring semivolatile $PM_{2.5}$ nitrate and OC than both the RAMS and CAMM. The regression between the differential TEOM $PM_{2.5}$ mass concentrations (Y) and the ICVC $PM_{2.5}$ nitrate concentrations (X) is $Y = 3.09(\pm 0.17)X + 10.8(\pm 1.24)$, $r^2 = 0.62 (n = 191)$. A poorer correlation was observed with the RAMS, having RAMS $= 1.37(\pm 0.19)NO_3^- +$

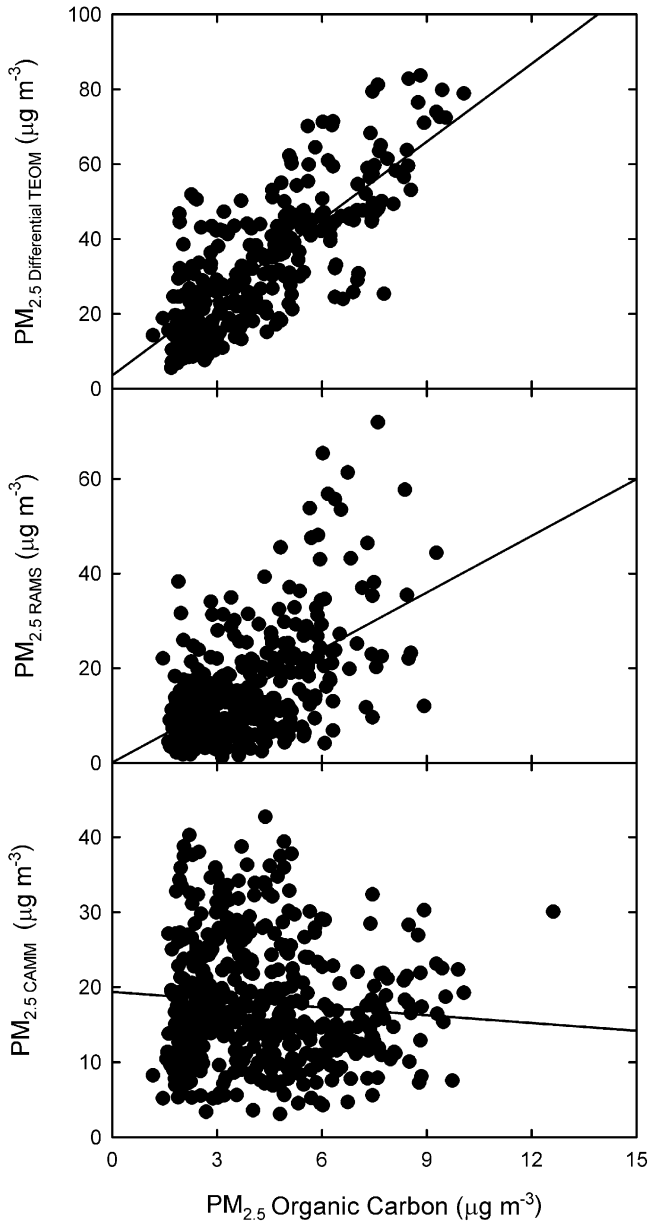


Fig. 4. Correlation between the organic carbon concentrations and $PM_{2.5}$ mass concentrations determined from the differential TEOM, the RAMS, and the CAMM.

7.07($\pm$1.31) with a low r^2 value of 0.20 ($n = 212$). No significant relationship was found between the hourly average concentrations of the CAMM $PM_{2.5}$ mass and the ICVC $PM_{2.5}$ nitrate concentrations.

A similar relationship was observed between the continuous $PM_{2.5}$ mass concentrations and fine particle OC by the Sunset Lab OC/EC analyzer (Fig. 4). The regression equation is $PM_{2.5}$ mass by the differential

TEOM $= 6.94(\pm 0.32)\ NO_3^- + 3.51(\pm 1.48)$, $r^2 = 0.61(n = 304)$. Higher slope and lower intercept are calculated for OC, compared to nitrate, with the same degree of correlation. Similar to the RAMS and fine particle nitrate, a relatively low correlation ($r^2 = 0.30$) was achieved between the RAMS and fine particle OC, along with the equation RAMS $= 3.99(\pm 0.33)$OC $+ 0.11(\pm 1.39)(n = 343)$. No significant relationship was found between the CAMM $PM_{2.5}$ mass and OC concentrations.

These results suggest that we should consider the degree of contribution by particulate nitrate and OC to the differential TEOM $PM_{2.5}$ mass. A different relationship can be seen between the PM mass and particulate species by looking at the data separated into periods of August and September. During the period from 15 August to 31 August, the correlation (r^2) between the paired differential TEOM $PM_{2.5}$ mass and the ICVC nitrate concentrations was as low as 0.38 ($n = 52$), while it increased up to 0.84 ($n = 138$) for the period from 1 September to 7 September. There were many missing nitrate data in the former period of time. Different results were shown between the differential TEOM PM mass and OC concentrations. The correlation (r^2) during the former time period was as relatively strong as 0.68 ($n = 168$), while a weak correlation of 0.30 ($n = 135$) was observed in the latter period. There were OC concentration data over the whole period of time. These results suggest that semivolatile organic aerosol also contributed to the differential TEOM $PM_{2.5}$ mass concentrations along with the particulate nitrate. However, the contribution by OC appeared higher during the period before August 31, while the contribution by nitrate appeared higher during the period after 1 September.

These results demonstrate that the differential TEOM measured $PM_{2.5}$ mass concentrations with lower losses of semivolatile nitrate and organic carbon than either the RAMS or the CAMM, and also the RAMS had lower losses than the CAMM, as seen in Figs. 3 and 4. This finding also provides an important argument that short term intervals (e.g., less than1 h) of sampling may not guarantee the measurement of reliable $PM_{2.5}$ mass concentrations against the loss of semivolatile species from particles collected, especially in areas with high semivolatile concentrations, unless appropriate devices are equipped to minimize volatilization of semivolatiles from, or adsorption of gas species onto, particles. Therefore, an appropriate selection of PM mass monitors should be made to collect "real" fine particle mass concentrations.

3.2. *Effects of semivolatiles on* $PM_{2.5}$ *mass*

The differential TEOM measures total (volatile and nonvolatile) $PM_{2.5}$ mass concentrations when the ESP is off, while particles are not collected for the periods when the ESP is energized. However, volatilization of previously collected particles and/or any adsorption of gaseous species in the sample stream may continue while the ESP is on. In this case, if volatilization of previously collected particles occurs, the PM mass concentrations are observed to be below zero. On the other hand, the PM mass concentrations above zero indicates that there was adsorption of gaseous species onto particle surface. The net ambient $PM_{2.5}$ mass concentration is then determined by subtracting the PM mass collected while the ESP is on from the PM mass collected while the ESP is off (Jaques et al., 2004).

Fig. 5 depicts moving 1 h averages of the 10 min ICVC nitrate concentrations and the differential TEOM $PM_{2.5}$ mass concentrations on 3 and 6 September, 2001 when the nitrate concentrations were relatively low and high, respectively. For the periods when the ESP is turned on, PM mass concentrations both above and below zero were observed, suggesting that both adsorption and volatilization occurred. However, at low nitrate concentration hours on 3 September, adsorption of gaseous species onto the filter and/or particle surfaces seems to contribute as much as 20 $\mu g/m^3$. Volatilization ($> 20\ \mu g/m^3$) of $PM_{2.5}$ mass occurred at hours of high nitrate concentration on 6 September. These results suggest that

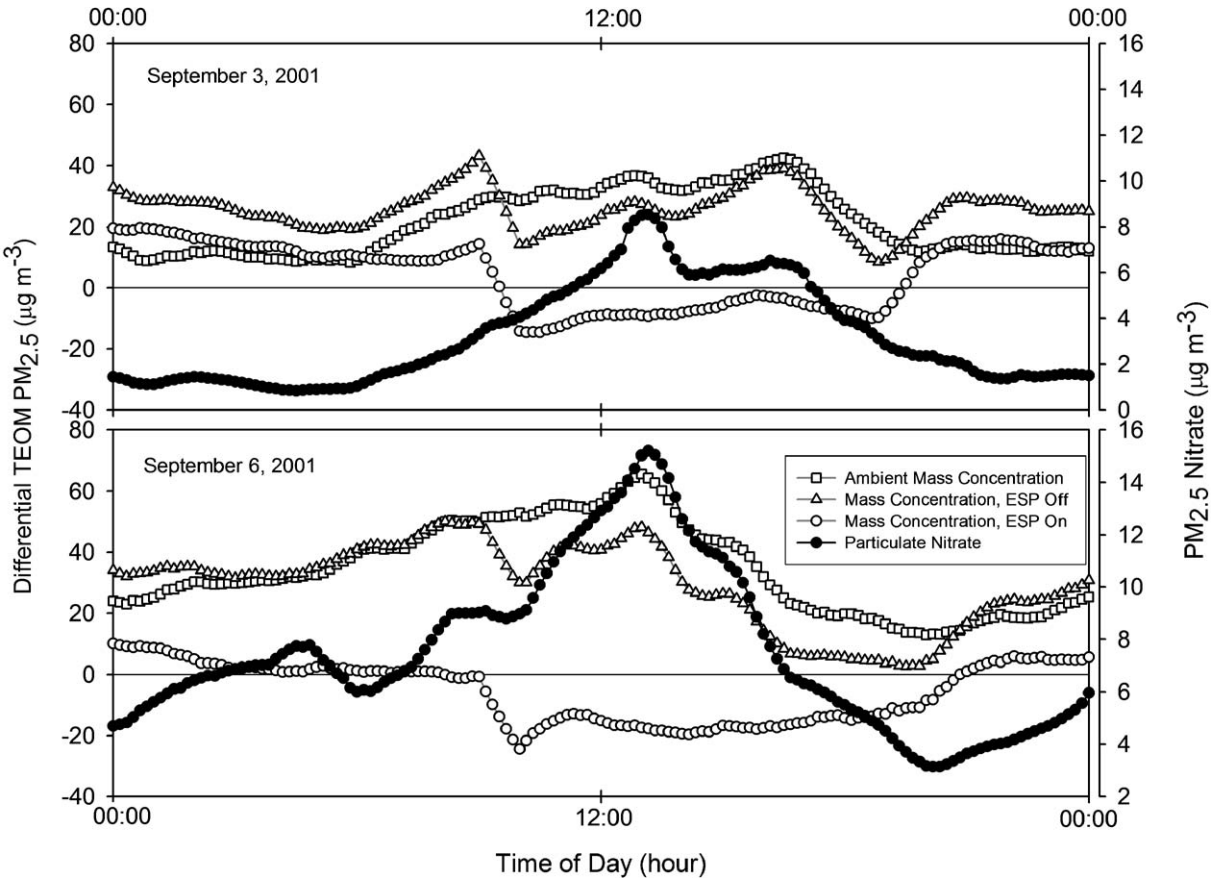


Fig. 5. Diurnal variations of the 10-min particulate nitrate and $PM_{2.5}$ mass concentrations as moving 1-h averages. The PM mass concentrations presented are from the differential TEOM, with ESP on/off data.

volatilization of $PM_{2.5}$ mass becomes substantial for the periods of high concentrations of nitrate while gas adsorption becomes more significant rather than volatilization effects when nitrate concentrations decreased. Note that the nitrate concentrations peaked during the daytime, indicating the formation of ammonium nitrate aerosol. In addition, due to high temperature and low water vapor in the air, particulate ammonium nitrate collected on the filter may be dissociated into gaseous ammonia and nitric acid, which may result in the reduction of $PM_{2.5}$ mass measured on the ESP on side. On the other hand, it is considered that volatilization of particulate nitrate occurred from previously collected particles on the differential TEOM filter.

Fig. 6 shows moving 4 h averages of the hourly OC concentrations and hourly differential TEOM mass concentrations from 22 August to 25 August, 2001. There were also volatilization and adsorption processes found as a function of the change of OC concentrations. As in the case of nitrate, volatilization of $PM_{2.5}$ mass was obvious for the periods of increasing OC concentrations. As suggested in the Fig. 6, the OC concentrations were likely to be associated with volatilization of previously collected particles, with a

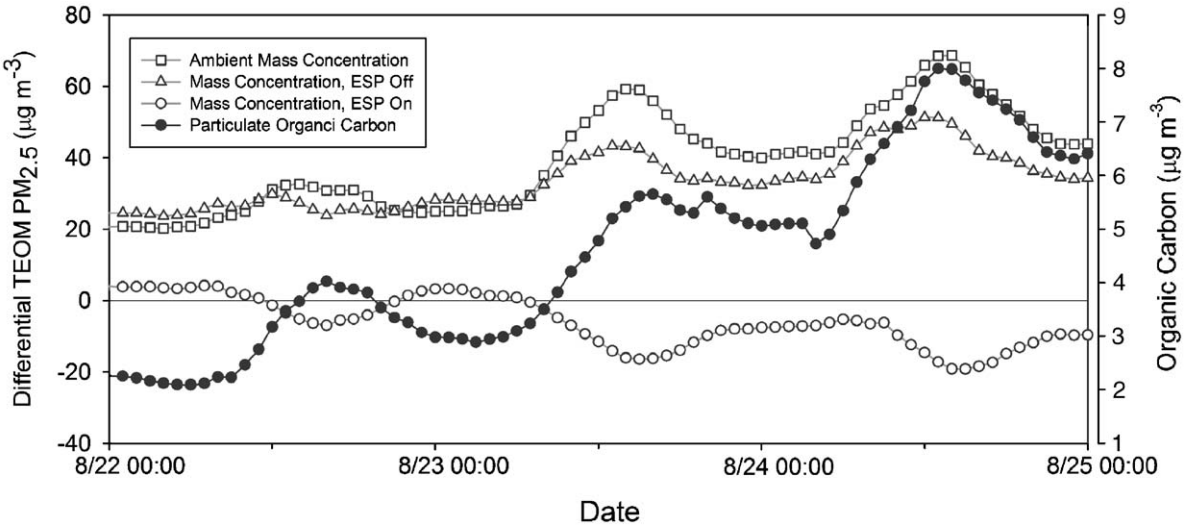


Fig. 6. Time series of particulate organic carbon and $PM_{2.5}$ mass concentrations as moving 4-h averages of the hourly concentrations, respectively. The PM mass concentrations from the differential TEOM with ESP on/off data are presented.

few exceptions. These results suggest that particulate OC may be another major contributor to evaporation of semivolatiles from the differential TEOM $PM_{2.5}$ mass. Volatilization of OC for the September period was not significant.

3.3. *Continuous to filter-based* $PM_{2.5}$ *mass*

The $PM_{2.5}$ mass concentrations of the differential TEOM were compared to those of the (Model 2025, Rupprecht and Patashnick, Co.) FRM sampler. The $PM_{2.5}$ mass concentrations of the differential TEOM were calculated, following the sampling intervals (4 to 10 h) of the FRM, and plotted for those concentrations of the partisol FRM sampler in Fig. 7. The linear equations show moderately high correlation ($r^2 = 0.52$) between $PM_{2.5}$ mass concentrations of the two samplers. The slope (0.91) of the linear fit equation is close to unity, but the intercept (11.3) is relatively high, indicating that higher PM mass concentrations were observed with the differential TEOM than with the FRM sampler. In fact, the 24-h averaged $PM_{2.5}$ mass concentrations were 26.8 and 36.4 $\mu g/m^3$ for the FRM and the differential TEOM, respectively. The average mass difference between the two samplers is 9.6 $\mu g/m^3$. This difference may be due to the volatilization of semivolatile nitrate from the PM mass collected on the FRM filter, presumably because none of the devices minimize evaporation of semivolatiles, the effect of longer sampling intervals, or both. No meaningful relationship was observed between either the CAMM or the RAMS and the FRM $PM_{2.5}$ mass concentrations (not shown) with r^2 values of 0.03 and 0.00, respectively.

To further support this hypothesis, the $PM_{2.5}$ mass differences between the paired differential TEOM and FRM were plotted against the particulate ammonium nitrate differences between the paired HEADS and FRM sampler (see Fig. 8). Here, assuming that all nitrate measured by the FRM and the HEADS is associated with ammonium nitrate, the nitrate concentrations were multiplied by 1.29 (i.e., the ratio

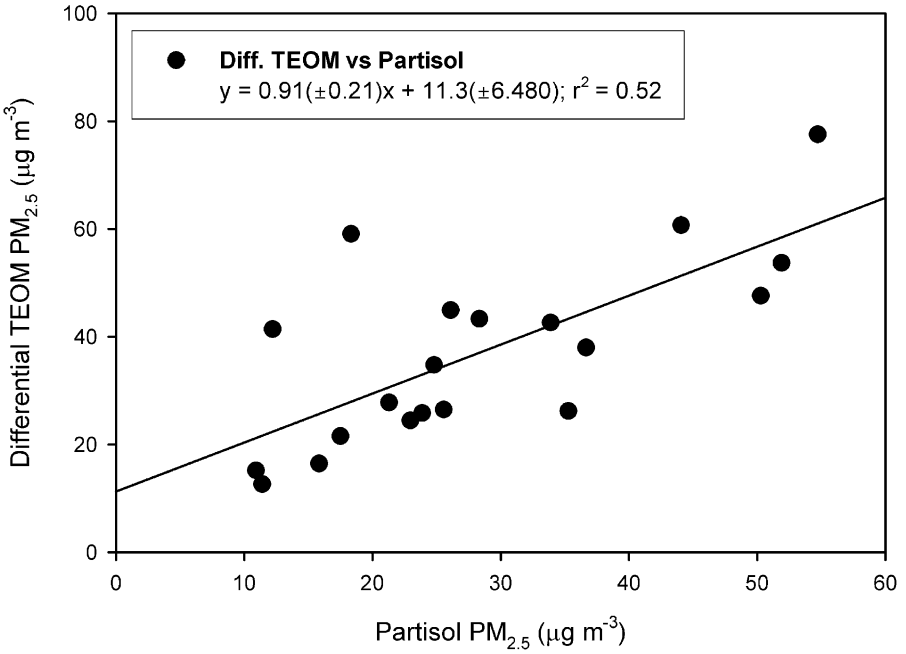


Fig. 7. Linearity between 24-h average $PM_{2.5}$ mass concentrations measured by the differential TEOM and the partisol FRM for full period from 17 August to 7 September, 2001.

of the molecular weights of ammonium nitrate and nitrate). For the 15 paired differences, the "mass loss" has a relatively high correlation with the "nitrate loss" ($r^2 = 0.51$). The relatively high correlation between the differential TEOM-FRM mass differences and the HEADS-FRM particulate ammonium nitrate differences suggests that there may be a significant relationship between the mass loss and nitrate loss from the partisol filter. The slope of 1.41 suggests that a substantial portion of mass loss may result from the volatilization of particulate ammonium nitrate, unlike the previous study conducted in Claremont, CA (Jaques et al., 2004). Jaques et al. concluded that the volatilization of semivolatile organics was considered to be as substantial in the loss of $PM_{2.5}$ mass as semivolatile ammonium nitrate. To strengthen this argument, a correlation between the paired differential TEOM and partisol mass differences and the continuous OC concentrations from the Sunset monitor was taken into consideration. The calculated correlation (r^2) was 0.17, suggesting that volatilization of semivolatile organic aerosol was relatively low, unlike as expected in Fig. 6, and thus the PM mass loss could be attributed to the semivolatile ammonium nitrate loss.

For the comparison with 24 h average concentrations of $PM_{2.5}$ mass, particulate nitrate concentrations from the partisol and HEADS were used. The nitrate concentrations were measured four times per day and converted to 24 h average concentrations, considering sampling air volume of each measurement time period, respectively. Here, the particulate nitrate is assumed to be particulate ammonium nitrate. The 24 h average particulate ammonium nitrate concentrations of the partisol and HEADS were 7.84 and 19.4 $\mu g/m^3$, respectively. The ammonium nitrate difference between the partisol and HEADS is 11.5 $\mu g/m^3$, on average, which is close to 9.6 $\mu g/m^3$, the mass difference between the paired differential TEOM and partisol. This result also indicates that the nitrate loss from the partisol FRM substrate may

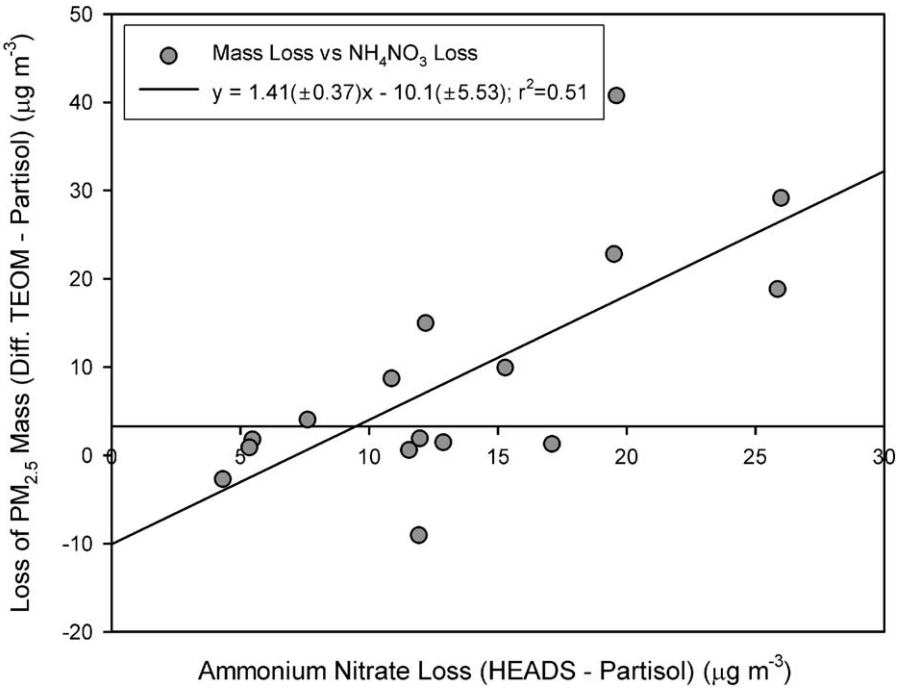


Fig. 8. Scatter plot presenting the relationship of the mass and nitrate losses by partisol with the differential TEOM and HEADS as reference.

make a substantial contribution to the mass loss between the differential TEOM and the partisol FRM sampler.

A statistical relationship between the differential TEOM and either the CAMM or the RAMS was also examined. However, no significant relationships were determined because those concentrations were distributed randomly.

4. Conclusions

The differential TEOM monitor showed better performances in the measurement of $PM_{2.5}$ mass concentrations, compared to the CAMM and the RAMS, by capturing fine particles with fewer losses of semivolatile ammonium nitrate and organic aerosol from filter. It has been shown that particulate ammonium nitrate made a large contribution to the Rubidoux fine particle mass, and thus it is important to measure $PM_{2.5}$ mass concentrations without losing nitrate during sampling. Therefore, it is necessary to employ a sampler with an appropriate device so that semivolatile nitrate evaporated from previously collected particles can be captured, since a short time interval of sampling may not be sufficient in the measurement of "real" ambient PM mass concentrations, especially in areas of high nitrate concentrations like Rubidoux. The study results suggest that $PM_{2.5}$ from the partisol FRM underwent a substantial mass loss of semivolatile ammonium nitrate, due to volatilization from filter, which resulted in the mass loss of $PM_{2.5}$ in the measurement of the FRM $PM_{2.5}$ mass concentrations.

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